Arjuna JEE 2025

Mathematics

Binomial Theorem

DPP: 1

Q1 If
$(1+ax)^n = 1+6x+\frac{27}{2}x^2+.....+a^n x^n$,
then the value of a and n are respectively
(A) 2,3
(B) 3,2
(C) $\frac{3}{2}, 4$
(D) 1,6

Q2 The number $(2+\sqrt{3})^6$ of can be expressed in the form $k+\sqrt{k^2-1}$ where k is a positive integer, then $k=$ (where $[\cdot]$ denotes greatest integer function)

Q3 The value of $\left\{(\sqrt{2}+1)^5+(\sqrt{2}-1)^5\right\}$, is
(A) 58
(B) $58\sqrt{2}$
(C) 42
(D) $42\sqrt{2}$

Q4 If n is a positive integer, then
$(\sqrt{3}+1)^{2n}-(\sqrt{3}-1)^{2n}$ is
(A) an irrational number
(B) an odd positive integer
(C) an even positive integer
(D) a rational number other than positive integers

Q5 If 6^{th} term in the expansion of
$\left(\frac{1}{x^{8/3}}+x^2\log_{10}^{x}\right)^8$ is 5600 then x is equal to
(A) 8
(B) 9
(C) 10
(D) None of these

Q6 Find
(i) 28 th term of $(5x+8y)^{30}$
(ii) 7 th term of $\left(\frac{4x}{5}-\frac{5}{2x}\right)^9$

Q7 If the second term in the expansion of $\left(\sqrt[13]{a}+\frac{a}{\sqrt{a^{-1}}}\right)^n$ is $14a^{5/2}$, then the value of $\frac{{}^nC_3}{{}^nC_2}$, is
(A) 4
(B) 3
(C) 12
(D) 6

Q8 If the co-efficient of x^7 and x^8 in the expansion of $\left(2+\frac{x}{3}\right)^n$ are equal, then the value of n is equal to

Q9 If the fourth term in the expansion of $\left(x+x^{\log_2 x}\right)^7$ is 4480 , then the value of x where $x\in N$ is equal to:
(A) 3
(B) 1
(C) 4
(D) 2

Q10 If the fourth term in the binomial expansion of $\left(\sqrt{\frac{1}{x^{1+\log_{10} x}}}+x^{1/12}\right)^6$ is equal to 200 , and $x>1$, then the value of x is
(A) 10
(B) 10^3
(C) 100
(D) 10^4

Q11 If $\left(\frac{3^6}{4^4}\right)k$ is the term, independent of x, in the binomial expansion of $\left(\frac{x}{4}-\frac{12}{x^2}\right)^{12}$, then k is equal to

Q12 In the binomial expansion of $(1+a)^{m+n}$, if the coefficient of a^m and a^n are A and B respectively, then
(A) $A=B$
(B) $mA=nB$

(C) $nA = mB$ (D) $A = 2B$

Q13 If A and B denote the coefficients of x^n in the binomial expansions of $(1+x)^{2n}$ and $(1+x)^{2n-1}$ respectively, then

(A) $A = B$ (B) $2A = B$

(C) $A = 2B$ (D) none of these

Q14 In the expansion of $\left(x^3 - \frac{1}{x^2}\right)^{15}$, the constant term,is

(A) ${}^{15}\mathrm{C}_9$

(B) 0

(C) $-{}^{15}\mathrm{C}_9$

(D) 1

Q15 The coefficient of x^4 in the expansion of $\left(\frac{x}{2} - \frac{3}{x^2}\right)^{10}$, is

(A) $\frac{405}{256}$

(B) $\frac{504}{259}$

(C) $\frac{450}{263}$

(D) none of these

Q16 If the coefficients of x^{-2} and x^{-4} in the expansion of $\left(x^{1/3} + \frac{1}{2x^{1/3}}\right)^{18}$, $(x > 0)$, are m and n respectively, then m/n is equal to

(A) $5/4$ (B) $4/5$

(C) 27 (D) 182

Q17 Find the term independent of x in $\left(\frac{3}{2}x^2 - \frac{1}{3x}\right)^9$

Q18 Find the coefficients of $x^{32} \& x^{-17}$ in $\left(x^4 - \frac{1}{x^3}\right)^{15}$

Answer Key

Q1 (C)

Q2 1351

Q3 (B)

Q4 (A)

Q5 (C)

Q6 i) $\frac{30!}{27!3!}5^3x^3 8^{27}y^{27}$

ii) $\frac{10500}{x^3}$

Q7 (A)

Q8 55

Q9 (D)

Q10 (A)

Q11 55

Q12 (A)

Q13 (C)

Q14 (C)

Q15 (A)

Q16 (D)

Q17 $\frac{7}{18}$

Q18 1365 and -1365